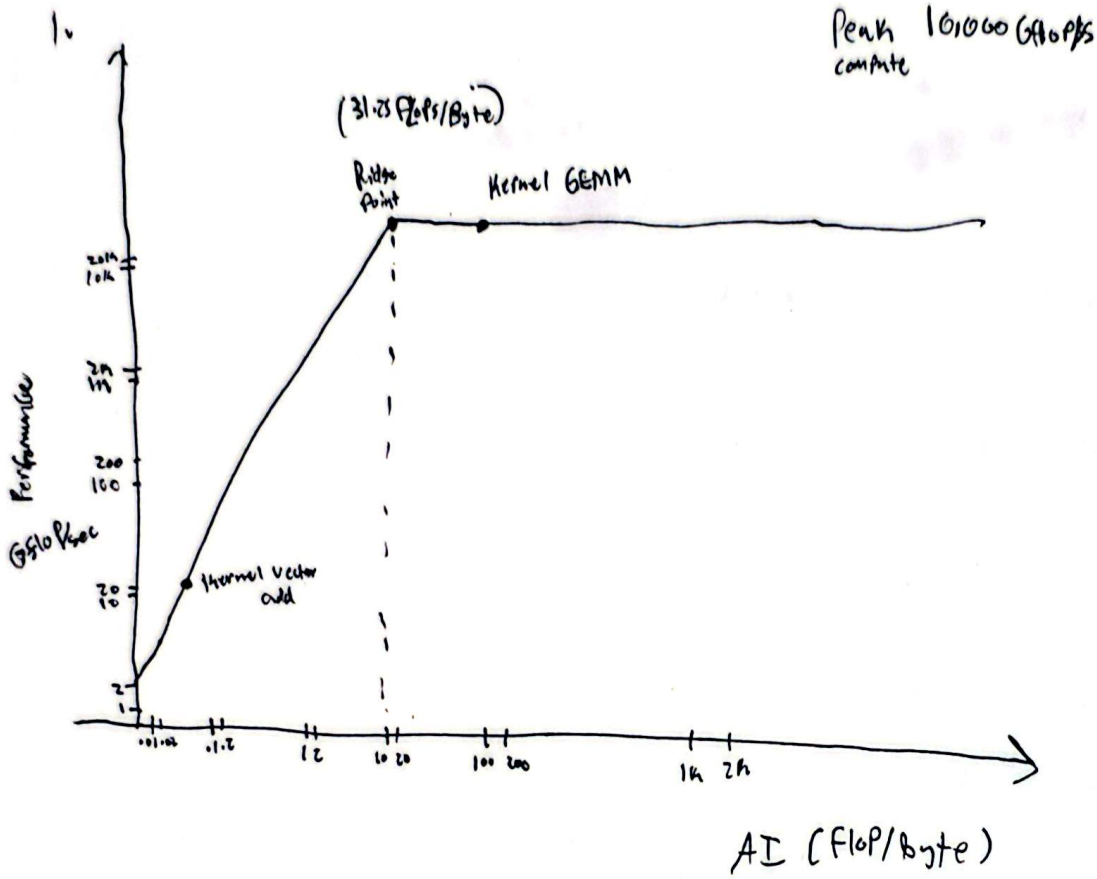
CMAN

1.



2. Given:

$N=1024$

Formula:

$2xN^3$

$Flops = 2x(1024)^3 = 2147,483,648$ Flops

Now bytes transferred!

Three matrices with N^2 elements:

$3x1024^2 * 4$ bytes/element $=$ 12,582,912 bytes

AI: 2,147,483,648/12,582,912 $\approx$ 170.67 Flop/byte

3. we know that
each vector is 4,194,304 long

Flops: This would be the same
as $N = 4{,}194{,}304$ flops

Bytes transferred:
$2 \cdot N = 2 \cdot 4194304 \cdot 4\text{bytes} = 33{,}554{,}432$ bytes

AI: $4{,}194{,}304 / 33{,}554{,}432 = 0.125$ flop/byte

4. 1) a) Compute Bound

b) 16,000 GFlop/s

c) Since it is compute bound, we need to increase processing power or throughput in some way. We can either have additional ALU, try to apply more parallelism or simply speed up the processor

2) a) memory bound

b) 0.125 flops/byte $\cdot$ 320 GB/s $= 40$ GFlop/s

c) Increase memory bandwidth in some way.